

Highlights

Breaking the Observation Ceiling: Conformal Fusion of Trace Signals for Detecting Prompt-Injection Compromise in LLM Agents

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- Fuses observation- and action-side trace signals to detect agent compromise
- Conformal surprisal fusion is scale-free and cannot clip out a weak signal
- Attains 0.960 TPR at 3.66% FPR, above the best published trace-only result
- 7.8% of successful injections leave no trace evidence, capping all monitors
- AgentDojo’s security flag is easily inverted; a ground-truth check is given